

Mixed-Length Shapelets to KMeans

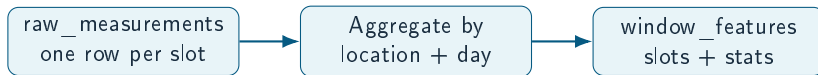
Picoclimate Example from Raw Data

Explainable Clustering for Spatio-Temporal Data

An Pham | LIRMM-ADVANCE | Supervisors: Luc & Reda

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Raw Measurements to Window Features



Slot rows are aggregated into one daily window per location.

Example Row: raw_measurements.csv (Truncated)

timestamp	location	slot	air_temp	humidity	wind	pressure
2025-09-01 06:00	loc_000	morning	50.0	5.0	17.226	969.0

- Each row is one location + one time slot (morning/noon/afternoon/night).
- These rows become a 4-slot daily window in `window_features.csv`.

Example Row: window_features.csv (Truncated)

location	date	present	air_temp_s1	wind_s1	air_temp_s2	wind_s2	true_regime
loc_049	2025-09-24	0.9348	50.0	12.516	50.0	12.045	traffic_peak

- Slot columns (`__slot_1..__slot_4`) are the shapelet input.
- Mean/std/trend columns exist but are not used in the shapelet pipeline.

Shapelet Input Vector (Picoclimate)

- Build one 1D vector per window using only slot columns.
- Example (truncated):
- This is **not** a single-variable series.
- It is a flattened multivariate window.

```
 $x$  = [air_temp_s1, humidity_s1, wind_s1, ...,  
air_temp_s2, humidity_s2, wind_s2, ..., air_temp_s4,  
...]
```

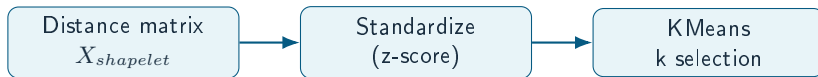
Mixed-Length Shapelets to Distance Features

- Candidate shapelets are sampled from mixed lengths (e.g., 5 and 10). Distance mapping
- Not all possible shapelets are enumerated; sampling is random.
- Each shapelet yields one feature: minimum distance to the window.

$$\phi_j(x) = \min_t \|x[t : t + L_j] - s_j\|_2$$

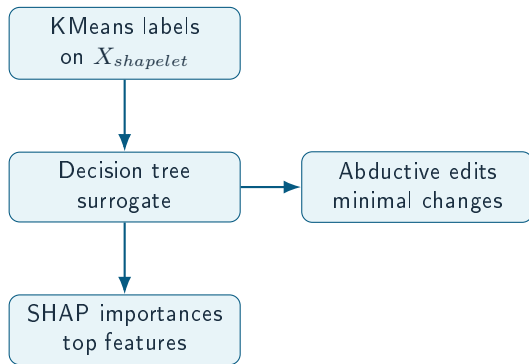
$$X_{shapelet} \in \mathbb{R}^{N \times m}$$

KMeans Input and Selection



Input to KMeans is the fixed-width shapelet-distance matrix.

Explanation Stage (Transparent)



- Surrogate learns cluster rules in shapelet space.
- SHAP ranks which shapelet distances drive the label.
- Abductive edits show minimal changes to switch cluster.

Clarifying the Question

- The current pipeline uses **all slot columns**, not one variable only.
- Mixed-length shapelets are sampled from the flattened slot vector.
- Distances become a fixed-width vector; that is the KMeans input.
- Using a single variable is possible, but changes the representation.